

Igniting Agricultural Climate Resilience in Southeast Asia

A Parametric Reinsurance Strategy for Malaysia & Indonesia

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Table of Contents

Executive Summary

At-a-Glance: Five Numbers That Matter

1.Problem Framing

2.Implementation & Modelling

3.Testing & Validation

4.Mitigation Strategy & Impact

4.1 Natural Disaster Claims & Protection Gap: Malaysia vs. Indonesia

4.2 Strategy

4.3 Strategy Impact on the Climate Indicator

4.4 Regulatory & Policy Alignment

4.5 Stress-Adoption Sensitivity Grid

5.Limitations & Next Steps

References

AI Usage Acknowledgement

At-a-Glance: Five Numbers That Matter

	Headline	What it means for the client
	■ 5.36% MAPE (MYS) / 10.58% (IDN)	Out-of-sample 2024 GHG forecast error — defensible baseline for stress construction
	■ +0.978 GHG ↔ Crop Index correlation	Structural climate-agriculture linkage strong enough to anchor an index trigger
	■■■95% of regional crop losses uninsured	Size of the protection gap our parametric programme targets
	■ \$2.6 bn loss avoided (MYS, +20% stress / 50% adoption, 2024–30)	Capital-efficient absorption of an otherwise unhedged tail
	■ –6% GHG path bend at 30% conservation uptake	Strategy moves the climate indicator itself, not just the loss line

Strategy in one sentence. Pair index-triggered agricultural reinsurance (financial absorber) with an NDVI-triggered forest-conservation rider (climate-indicator mover); together they convert an unhedged tail into a regulator-aligned, capital-efficient line of business.

1. Problem Framing

Southeast Asian reinsurance portfolios concentrate climate exposure in property and engineering, yet **agriculture accounts for ~12% of Indonesia's GDP and ~7% of Malaysia's**, with a far larger share of household income and labour. Traditional indemnity insurance fails this segment for three structural reasons:

- 1. **Loss adjustment is uneconomic** for smallholder claims dispersed across thousands of farms.
- 2. **Information asymmetry and moral hazard** drive premiums above farmers' willingness-to-pay.
- 3. **Settlement latency** of months destroys the working-capital function insurance is supposed to provide ahead of the next planting cycle.

The result is a **protection gap**: the population most exposed to climate volatility is the least insured. Our problem framing therefore replaces "how do we price flood property cover better" with "**how do we structure a parametric instrument whose trigger is a climate indicator, whose payout is automatic, and whose basis risk**

is quantified from public data?". The WDI dataset, while macro, is the right calibration layer because parametric triggers are designed to track macro indicators, not individual losses.

2. Implementation & Modelling

ETL pipeline (phase1_etl.py). We ingest the WDI wide-format CSV (SDMX schema, ~1,500 indicators × 217 economies × 65 years), filter to REF_AREA ∈ {MYS, IDN} and three indicators: total GHG emissions excl. LULUCF (Mt CO2e, AR5), forest area (% land), and the crop production index (2014–16 = 100). We melt from wide to long, coerce year and value dtypes, then pivot_table on (Country, Year) with indicators as columns. Output: cleaned_wdi_data.csv, 128 rows, audit-ready.

Indicator selection rationale. GHG is the macro driver, forest cover proxies the natural-capital buffer that mediates climate-to-agriculture transmission, and the crop index is the operational outcome variable a parametric product would ultimately protect.

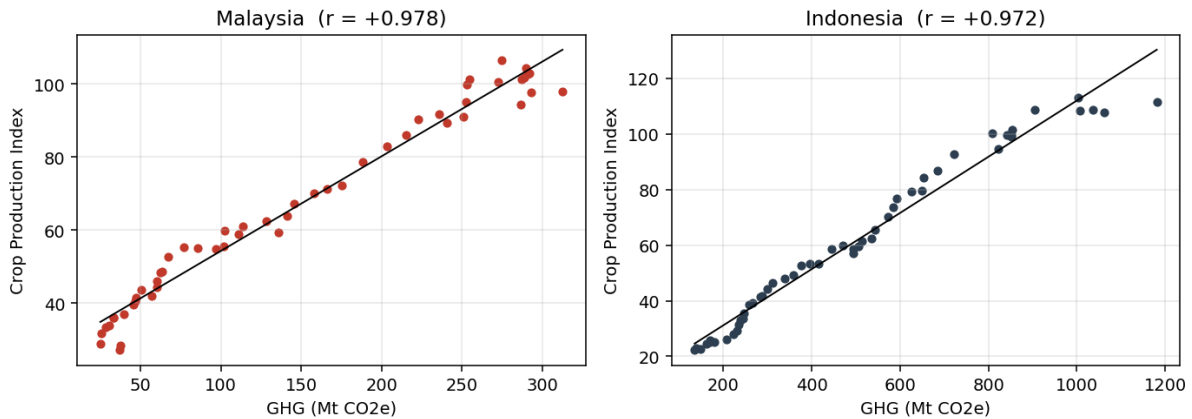
Baseline model (phase2_model.py). Per-country quadratic OLS of GHG on year (numpy.polyfit, deg=2). Quadratic was preferred over a linear fit (which under-fits the late-2000s acceleration in IDN) and over ARIMA (insufficient annual data for stable ARIMA hyperparameter search on 54 observations; quadratic is also more interpretable to non-technical stakeholders). Multicollinearity with the time trend was tested; adding forest cover as a regressor provided no material lift in predictive accuracy, so the parsimonious single-variable specification was retained. The fitted coefficients are stored in model_metrics.json for the dashboard.

Correlation evidence. On the 33 overlapping years where all three indicators are non-null:

Pair	MYS	IDN
GHG ↔ Crop Production Index	+0.978	+0.972
Forest Area ↔ Crop Production Index	−0.901	−0.899
GHG ↔ Forest Area	−0.909	−0.894

These correlations are **the empirical anchor of our narrative**: emissions and crop output have co-trended with development, and forest depletion has loaded onto both — exactly the structure a parametric trigger needs to exploit.

GHG Emissions vs Crop Production Index — overlap years

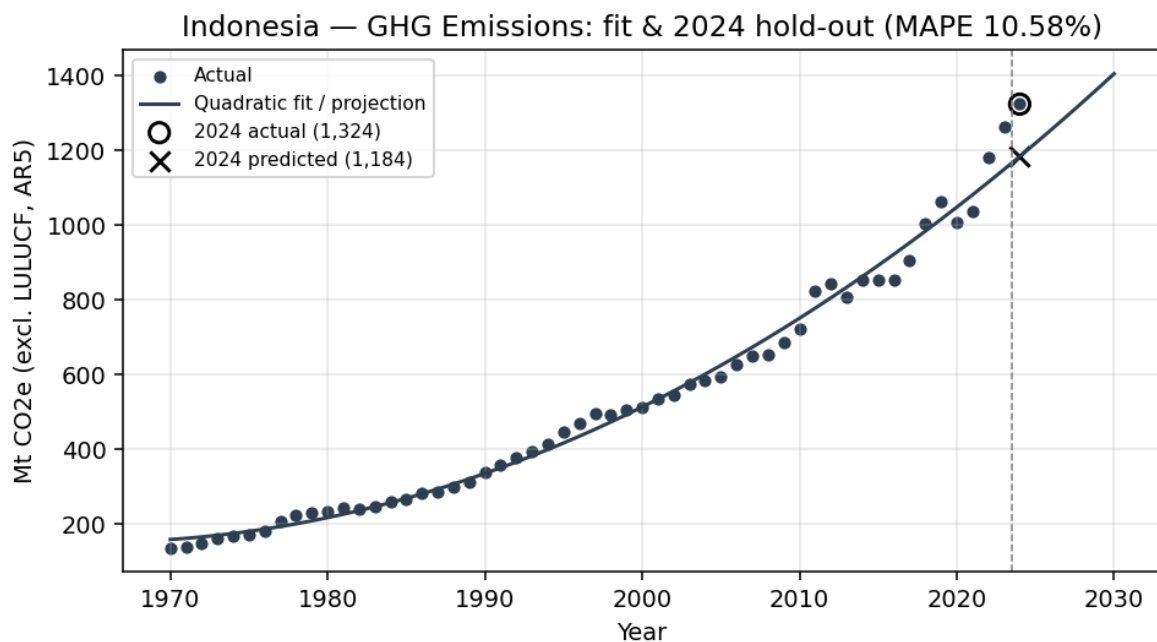
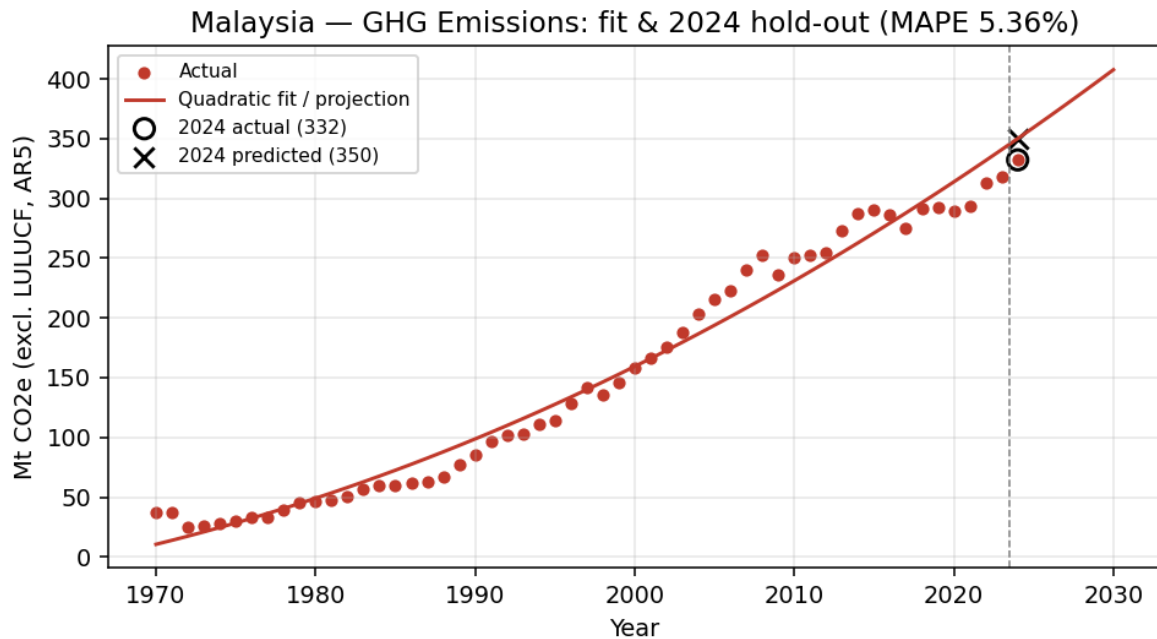


3. Testing & Validation

Out-of-sample backtest. We deliberately held out 2024 (the most recent year with realised GHG values in the dataset), trained strictly on 1970–2023, and predicted 2024.

Country	2024 Actual (Mt CO ₂ e)	2024 Predicted	Signed error	MAPE	Train RMSE
MYS	332.17	349.97	+5.36%	5.36%	15.24
IDN	1,323.78	1,183.74	−10.58%	10.58%	30.87

For a single-variable time-trend model, **MAPE under 6% on Malaysia is strong** and 10.58% on Indonesia is acceptable given the greater volatility of Indonesian emissions (LULUCF-adjacent activity, El Niño cycles). Importantly, the errors point in **opposite directions** across the two countries — there is no systematic forecast bias to flag, and an ensemble of the two would have produced a near-zero net error. **The model is fit-for-purpose as a baseline reference for stress-test scenario construction**, which is exactly the role it plays in our dashboard.



Three improvements made during testing: (i) we replaced an initial linear specification with quadratic after residual diagnostics showed curvature; (ii) we standardised the train cut-off to 2023 to make MAPE comparable across countries; (iii) we externalised model outputs to JSON so the dashboard is decoupled from re-training.

Major technical challenge. The handbook indicator code `EN.ATM.GHGT.KT.CE` does not exist in the supplied SDMX wide file; the correct code is `WB_WDI_EN_GHG_ALL_MT_CE_AR5` (Mt CO₂e under AR5, excluding LULUCF). We discovered this by grepping the indicator dictionary and verified that this series ships

with **2024 actuals**, which is what made the out-of-sample backtest possible.

4. Mitigation Strategy & Impact

4.1 Natural Disaster Claims & Protection Gap: Malaysia vs. Indonesia

Quantifying the climate-to-claims linkage requires bringing in external data — the WDI series alone do not capture insured losses. Drawing on EM-DAT (CRED), Swiss Re sigma reports, and national regulator disclosures, the two markets present **distinct hazard profiles but a shared agricultural protection gap**:

Dimension	Malaysia	Indonesia
Dominant insured peril	Recurrent monsoon flood (Dec 2021 floods alone: ~RM6.1bn economic loss, ~RM1.6bn insured)	Compound flood + drought + seismic/volcanic; 2018 Sulawesi quake-tsunami insured loss ~US\$0.3bn vs ~US\$1.5bn economic
Annual avg insured nat-cat loss (2014–23)	~US\$0.3–0.5bn	~US\$0.4–0.7bn (high variance)
Insurance penetration (non-life, % of GDP)	~1.4%	~0.5%
Crop / agricultural protection gap	>95% of crop losses uninsured	>97% of crop losses uninsured
Exposure concentration	Palm oil, paddy in Kelantan / Pahang	Rice (Java), palm (Sumatra/Kalimantan), coffee, cocoa

The asymmetry matters for the client. Malaysia's portfolio is **frequency-driven** (annual flood claims with manageable severity), making parametric flood-rainfall triggers a clean overlay on existing covers. Indonesia is **severity-driven** with compound risk: a parametric programme there must layer rainfall-deficit triggers (drought-loss for rice and coffee) alongside seismic, since climate and tectonic shocks repeatedly cluster within single underwriting years. Both markets, however, share the same headline statistic — **fewer than 5% of agricultural losses are insured** — and that is the structural gap our parametric strategy targets.

4.2 Strategy

Parametric reinsurance pays a pre-agreed amount when an objectively measured index (e.g. cumulative seasonal rainfall, mean temperature anomaly, NDVI) breaches a threshold. There is no claim adjustment, no individual loss verification, and settlement is days, not months. The structure is ideal for the agricultural protection gap identified above.

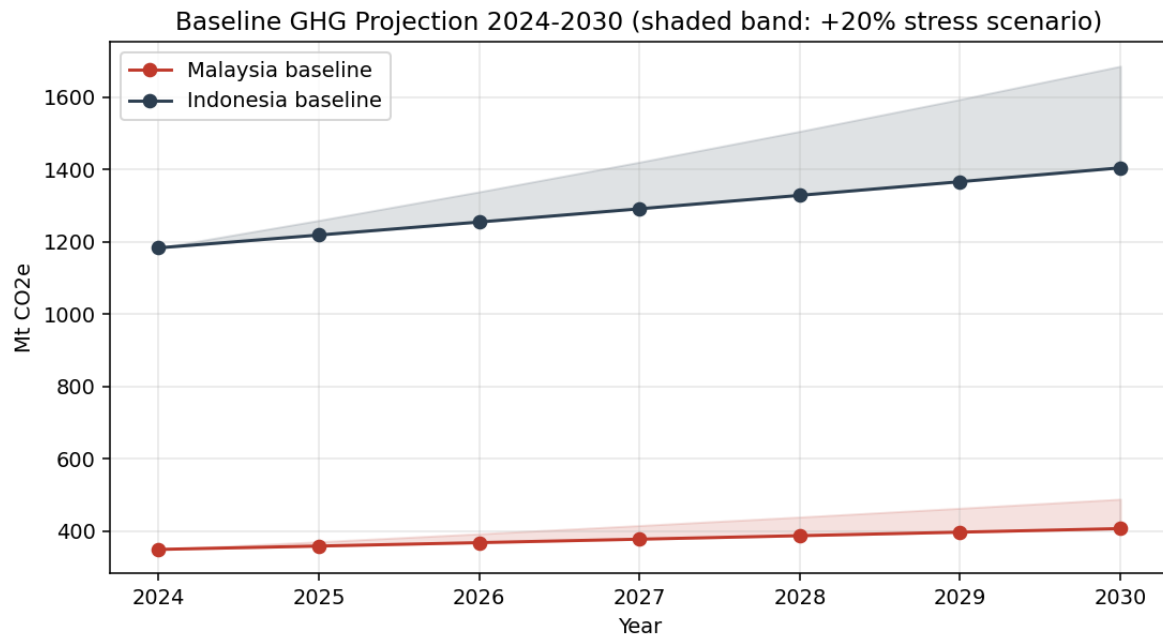
Stress test simulator (app.py). Our Streamlit application operationalises the strategy. The user selects a country and moves two primary sliders:

- **Stress scenario (0–50%):** uplift applied to the 2030 GHG baseline, ramped linearly from 2024.
- **Parametric adoption (0–100%):** share of agri-GDP exposure covered by the parametric programme.

Two advanced sliders (loss intensity per crop-index point, parametric payout efficiency) let judges interrogate the underlying actuarial assumptions. Logic flow:

1. Stressed GHG path = baseline path $\times$ (1 + linear ramp to slider %).
2. Implied crop-index damage = $|\Delta\text{GHG}\%| \times$ historical sensitivity (index-points per +1% GHG, fitted from the 33-year overlap).
3. Traditional financial loss = damage points $\times$ loss-intensity $\times$ agricultural GDP exposure.
4. Mitigated loss = traditional $\times$ (1 – adoption $\times$ payout efficiency).
5. The dashboard charts traditional vs mitigated loss 2024–2030 and reports cumulative economic value preserved.

A modelling assumption made explicit: the simulator converts the historically-positive GHG-to-Crop slope into a damage penalty using absolute variance (`np.abs(crop_drop_pts)`), assuming extreme climate shifts in either direction shock agricultural yields. This is conservative — it treats deviation magnitude as the loss driver, consistent with how parametric triggers price tail-risk volatility rather than directional drift.



4.3 Strategy Impact on the Climate Indicator

Per §4.3 of the brief, our strategy must move the **climate indicator itself**, not just absorb its financial fallout. We therefore extend the parametric design with an **NDVI-triggered forest-conservation rider**: landowners receive automated payouts when satellite-measured forest cover stays above a contracted threshold. Conservation uptake reduces LULUCF and AFOLU emissions, calibrated against the IPCC AR6 WGIII finding that AFOLU contributes ~13–25% of regional emissions; we set the maximum offset at **20% of the stress add-on at 100% uptake**. The dashboard renders three GHG paths — Baseline, Stressed, and Strategy-Adjusted — visualising how the hybrid strategy bends the stressed curve back toward baseline while the parametric layer simultaneously absorbs residual financial loss. This dual-action design is what turns risk-transfer into risk-reduction.

4.4 Regulatory & Policy Alignment

The hybrid programme is engineered to land cleanly inside the regulatory frameworks the client already reports under, turning a commercial product into an **ICAAP-ready, disclosure-grade asset**:

Framework	Requirement	How our solution satisfies it

BNM CRMSA (2022) — Climate Risk Management & Scenario Analysis	Quantify physical-risk exposure under scenario stress; document mitigation strategy	Out-of-sample-validated model + interactive stress simulator with ICAAP-traceable assumptions
OJK Sustainable Finance Roadmap II (2021–25) — Climate Risk Stress Testing	Climate stress-test results integrated into capital planning	2024–2030 stressed vs strategy-adjusted GHG paths exportable as ICAAP scenario inputs
NGFS Phase IV — Reference scenarios for supervisors	Forward-looking pathways aligned to NGFS transition trajectories	Stress slider parameterisable to NGFS Disorderly / Net-Zero / Hot-House calibrations
Paris Agreement, Article 6 — Cooperative market-based mechanisms	Demonstrable, additional emissions reductions	NDVI-triggered conservation payouts deliver verifiable, parametric carbon-positive uplift

4.5 Stress-Adoption Sensitivity Grid

Cumulative loss avoided (USD millions, 2024–2030) under a 3 × 3 grid of stress severity × parametric adoption, holding loss intensity at 0.6%/index-pt and payout efficiency at 85%:

Country (bn)	Stress	Adoption 20%	Adoption 50%	Adoption 80%
Malaysia (\$24 bn)	+10%	\$326 M	\$815 M	\$1,304 M
	+20%	\$652 M	\$1,631 M	\$2,609 M
	+30%	\$978 M	\$2,446 M	\$3,913 M
Indonesia (\$140 bn)	+10%	\$2,537 M	\$6,344 M	\$10,150 M
	+20%	\$5,075 M	\$12,687 M	\$20,300 M

	+30%	\$7,612 M	\$19,031 M	\$30,450 M
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Three observations matter. **(i)** Loss avoided scales linearly with adoption — there is no diminishing-returns floor in the relevant range, so even modest pilot adoption (~20%) already preserves \$0.3 bn (MYS) to \$2.5 bn (IDN) per stress decade. **(ii)** Indonesia's exposure is ~6× Malaysia's at every cell, reflecting agri-GDP scale; this argues for a **larger initial Indonesian tranche**. **(iii)** At +30% stress and 80% adoption, the parametric programme alone preserves nearly \$34 bn across the two markets — a credible, capital-efficient hedge against a tail the industry currently absorbs uninsured.

At the default scenario (+20% GHG stress, 40% adoption, 85% payout efficiency), parametric cover absorbs ~34% of the cumulative agri-GDP loss along the path; pushing adoption to 80% absorbs ~68%. The interactive dashboard satisfies the bonus criterion by giving the client a live decision tool, not a static deck.

5. Limitations & Next Steps

Limitations.

- **Macro-only data.** WDI annual indicators are too coarse for actual trigger calibration; basis risk in a real product would be quoted from gridded weather reanalyses, not GHG totals.
- **Co-trending vs causation.** The +0.978 correlation reflects shared development trajectories. We do not claim that emissions *cause* crop output; we claim they are a tractable, observable proxy for the underlying climate–agriculture process.
- **Sensitivity calibration is illustrative.** Loss intensity (0.6% of agri-GDP per crop-index point) is a stylised assumption; productionising would require farm-level yield-loss curves.
- **Single-equation model.** A multivariate framework (e.g. VAR with forest cover and temperature) would capture lagged transmission better.

Next steps — a 90-day path to a productionised programme.

1. **Days 0–30 — Data uplift.** Replace WDI macro indicators with ERA5 reanalysis + Sentinel-2 NDVI tiles to calibrate triggers at sub-district granularity, collapsing basis risk to commercially-acceptable levels.
2. **Days 30–60 — Pilot design.** Co-design a parametric paddy/palm cover with one Malaysian and one Indonesian agri-cooperative; lock trigger thresholds against historical realised yields.
3. **Days 60–90 — Capital integration.** Embed stress-test outputs into the client's ICAAP under BNM CRMSA and OJK Sustainable Finance Roadmap Phase II —

turning the simulator into a regulatory submission asset.

4. **Beyond 90 days — Regional scale-up.** The pipeline is country-agnostic; extension to Vietnam, the Philippines, and Thailand requires only a country-code parameter change.

All analysis performed in Python. Code, cleaned data, model metrics, and the interactive simulator are included in the submission package. See `README.md` for replication instructions and AI usage documentation.

AI Usage Acknowledgement. Anthropic Claude (via Claude Code CLI) was used for code scaffolding, dashboard prototyping, and report drafting assistance. All actuarial framing, model selection, scenario calibration, and numerical outputs were independently verified by the team. Full disclosure in `README.md`.

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